

```
SW1# show vlan brief
```

VLAN Name	Status	Ports
10 Administration	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5
20 Faculty	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9
30 Students	active	Fa0/10, Fa0/11, Fa0/12
40 Guests	active	Fa0/16, Fa0/17, Fa0/18

```
PC-Admin1> ping 192.168.10.2
```

```
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128 [Success]
```

```
PC-Admin1> ping 192.168.30.2
```

```
Reply from 192.168.30.1: Destination host unreachable. [Blocked by ACL]
```

```
PC-Student1> ping 192.168.20.2
```

```
Reply from 192.168.20.2: bytes=32 time=1ms TTL=127 [Success via R1]
```

```
R1# show ip route
```

```
Codes: C - connected, S - static, R - RIP, O - OSPF
```

```
Gateway of last resort is not set
```

```
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
C 192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
C 192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
C 192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
```

```
R1# show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gi0/1	trunk	802.1q	trunking	999

Port	Vlans allowed on trunk
Gi0/1	10,20,30,40

```
PC-Guest1 > Desktop > IP Configuration
```

```
IP Configuration:      ( ) Static  (*) DHCP
```

```
IPv4 Address .....: 192.168.40.23
```

```
Subnet Mask .....: 255.255.255.0
```

```
Default Gateway .....: 192.168.40.1
```

```
DNS Server .....: 8.8.8.8
```

```
DHCP request successful.
```

```
Lease obtained from R1 (GUEST_POOL).
```

```
R1(config)# interface gi0/0.10
```

```
R1(config-subif)# encapsulation dot1q 10
```

```
R1(config-subif)# ip address 192.168.10.1 255.255.255.0
```

```
R1(config-subif)# exit
```

```
R1(config)# interface gi0/0.20
```

```
R1(config-subif)# encapsulation dot1q 20
```

```
R1(config-subif)# ip address 192.168.20.1 255.255.255.0
```

```
R1(config-subif)# exit
```

```
R1(config)# interface gi0/0.30
```

```
R1(config-subif)# encapsulation dot1q 30
```

```
R1(config-subif)# ip address 192.168.30.1 255.255.255.0
```

```
R1(config-subif)# exit
```

```
R1(config)# interface gi0/0.40
```

```
R1(config-subif)# encapsulation dot1q 40
```

```
R1(config-subif)# ip address 192.168.40.1 255.255.255.0
```

```
R1(config-subif)# exit
```

```
R1(config)# interface gi0/0
```

```
R1(config-if)# no shutdown
```